Data Management and Sharing Plan

Types of Data and Products

Track 1 produces five classes of research products: (1) open-source software, including the existing FreeMoCap product, its component repositories, and the governance, documentation, and contributor-onboarding artifacts developed during the project; (2) ecosystem-discovery outputs — stakeholder findings, user and contributor barrier maps, and requirements documents; (3) validation and benchmark datasets, including reference recordings and cross-version regression baselines, that support publications; (4) educational and training materials developed in community pilots; and (5) minimal operational telemetry describing aggregate software usage.

Standards and Metadata

Software releases follow semantic versioning with tagged, documented releases. FreeMoCap recordings follow a documented, versioned folder structure that separates raw video, de-identified annotated video, and derived kinematic data; this taxonomy is a project standard, and it doubles as the mechanism by which contributors de-identify shared data (see Access and Sharing). In parallel with the security scoping described in the Project Description, the project will define provenance metadata sufficient to trace any produced dataset to the software versions, pose-estimation models, and processing parameters that generated it. Where community standards exist for new data types, they will be adopted; where they do not, formats and schemas will be documented and versioned publicly.

Access, Sharing, and Privacy

Privacy by architecture. FreeMoCap is a standalone desktop application in which all processing occurs on the user's own computer. Recordings of human participants — video of identifiable bodies foremost among them — never leave the user's machine, and the workflow is operable in fully air-gapped settings. The ecosystem shares software, derived kinematic data, and benchmarks; it is not a repository of user biospecimens.

Community-contributed data. Community members may voluntarily submit recordings for demos, showcases, and troubleshooting through a published media-sharing form that implements tiered, per-submission consent along two axes: data identifiability (raw video; de-identified annotated video; non-identifying 2D/3D kinematics) and sharing context (full public; limited public, e.g. the Discord community or conference presentations; internal use by the development team). The form includes self-service de-identification instructions, credit preferences, and a privacy notice describing each data type. Submissions are used and redistributed only within the selected tiers.

Telemetry and operational data. Telemetry consists of basic usage events keyed to a random, non-identifying per-installation UUID and is used to produce aggregate usage statistics. Raw telemetry logs are retained internally and are not shared.

Planned account systems. As the ecosystem develops credentialing and training-recording capabilities, user accounts will use standard OAuth2 authentication on Google Cloud Platform infrastructure, with minimal data collection, published disclosure, and user-facing controls. Payment processing is delegated to a commercial platform (Shopify); payment information is never handled by Foundation systems.

Exceptions to immediate sharing. Two classes of data will not be shared publicly, per the justification requirement of this plan: raw interview recordings and transcripts from ecosystem-discovery activities (human participant confidentiality; only aggregated findings will be published), and raw telemetry logs (personal privacy; only aggregates will be published).

Re-use and Redistribution

Software is licensed under the AGPL-3.0, so that improvements and extensions return to the community under the same terms. Benchmark and validation datasets will be released under permissive terms (CC-BY 4.0) to maximize re-use with attribution. Educational and training materials will be released under CC-BY 4.0. Governance and documentation artifacts are published in public repositories. Derivative works of community-contributed media are governed by the consent tier selected by the contributor at submission.

Archiving and Preservation

Tagged software releases are archived through the Zenodo–GitHub integration, yielding a citable DOI per release, so that publications can reference immutable, permanently available versions. Datasets supporting publications will be deposited in a DOI-issuing public repository at the time of publication, per NSF public access requirements. Documentation, governance artifacts, and educational materials are version-controlled in public repositories. The ecosystem's distributed development model provides additional resilience: complete copies of the product exist across many independent stakeholders and hosting locations.